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1 INTRODUCTION

This chapter introduces the domain of big data in cybersecurity, details the datasets used for the analysis, and outlines the primary objectives of the project.

1.1 Domain Introduction

The proliferation of internet-wide scanning has fundamentally changed the landscape of cybersecurity, providing both attackers and defenders with unprecedented visibility into the state of networked systems. As researchers like Durumeric et al. have shown, analyzing global scan data can reveal widespread vulnerabilities and coordinated attack campaigns [4]. However, the sheer volume of this data, which often scales to terabytes per day, presents a significant challenge for traditional analysis tools. This creates a critical gap in threat detection and response.

This project addresses this challenge by implementing a scalable data analysis model inspired by the objectives outlined in our initial synopsis. We leverage a distributed computing framework, Apache Spark, running on a Hadoop cluster, to automate the analysis of large-scale network datasets for the purpose of identifying and profiling malicious activities [5]. By implementing a multi-stage data analytics pipeline, this project demonstrates a practical approach to transforming raw network data into actionable cybersecurity intelligence, a critical step for modern Security Operations Centers (SOCs).

1.2 Datasets Used

Two distinct datasets were used to build and validate the threat profiling engine:

1. **CTU-13 Dataset [1]:** This dataset, consisting of botnet traffic from CTU University, is a well-established benchmark for botnet detection research. A subset of **300,000 records** was used to test the model's ability to differentiate between general 'ATTACK' and 'BENIGN' traffic profiles.
2. **CIC-Collection Dataset [2]:** This is a comprehensive and modern dataset from the Canadian Institute for Cybersecurity, featuring a wide array of network attacks. A substantial volume of **9,167,581 records** was processed to demonstrate the pipeline's scalability and its capability to profile specific, fine-grained attack types like 'DoS-Hulk' and 'DDoS-LOIC-HTTP' [3].

1.3 Purpose and Objectives

The primary purpose of this project is to develop an end-to-end, scalable pipeline for automated threat intelligence generation. The key **objectives** are:

- To implement a scalable data processing pipeline on a simulated Hadoop/Spark cluster to ingest and preprocess large-scale network traffic datasets [6].
- To build a threat behavior profiling engine using unsupervised machine learning (K-Means) to identify distinct attack patterns within known threat categories [7].
- To develop a recommendation engine using collaborative filtering (ALS) to automatically suggest appropriate mitigation strategies for each identified threat profile.
- To evaluate the stability and robustness of the generated profiles using Monte Carlo simulations and present the findings in a comprehensive, publication-ready report [8].

1.4 Dataset Characteristics

The feature importance analysis stage of the pipeline identified different key features as the most predictive for each dataset, reflecting their unique traffic compositions.

1.4.1 CTU-13 Dataset

For the CTU-13 dataset, the model identified features related to flow duration and packet counts as most important for distinguishing between generic attack and benign traffic.

Table 1: Key feature characteristics from the CTU-13 Dataset.

Feature Name	Description	Type	Role
flow_duration	Total duration of the flow in microseconds	Numeric	Feature
destination_port	The network port on the destination machine	Numeric	Feature
total_fwd_packets	Total packets sent in the forward direction	Numeric	Feature
Label	Traffic category ('ATTACK' or 'BENIGN')	Categorical	Target

1.4.2 CIC-collection Dataset

For the more complex CIC-collection dataset, the key features were related to TCP window sizes and packet length statistics, indicating more nuanced attack behaviors [3].

Table 2: Key feature characteristics from the CIC-collection Dataset.

Feature Name	Description	Type	Role
init_fwd_win_bytes	Initial TCP window size in forward direction	Numeric	Feature
fwd_packet_length_max	Max packet size in forward direction	Numeric	Feature
fwd_packet_length_std	Std. dev. of forward packet size	Numeric	Feature
bwd_packet_length_std	Std. dev. of backward packet size	Numeric	Feature
avg_fwd_segment_size	Average size of forward segments	Numeric	Feature
bwd_packet_length_mean	Mean size of backward packets	Numeric	Feature
flow_duration	Total duration of the flow in microseconds	Numeric	Feature
subflow_fwd_packets	Number of forward packets in a subflow	Numeric	Feature
packet_length_max	Maximum packet length in the flow	Numeric	Feature
idle_min	Minimum idle time in the flow	Numeric	Feature
Label	Specific attack type (e.g., 'DoS-Hulk')	Categorical	Target

2 METHODOLOGY

This chapter outlines the data analytics pipeline implemented in this project. The entire workflow was executed using PySpark, leveraging its distributed machine learning libraries (MLlib) and data processing capabilities, with data storage managed by the Hadoop Distributed File System (HDFS).

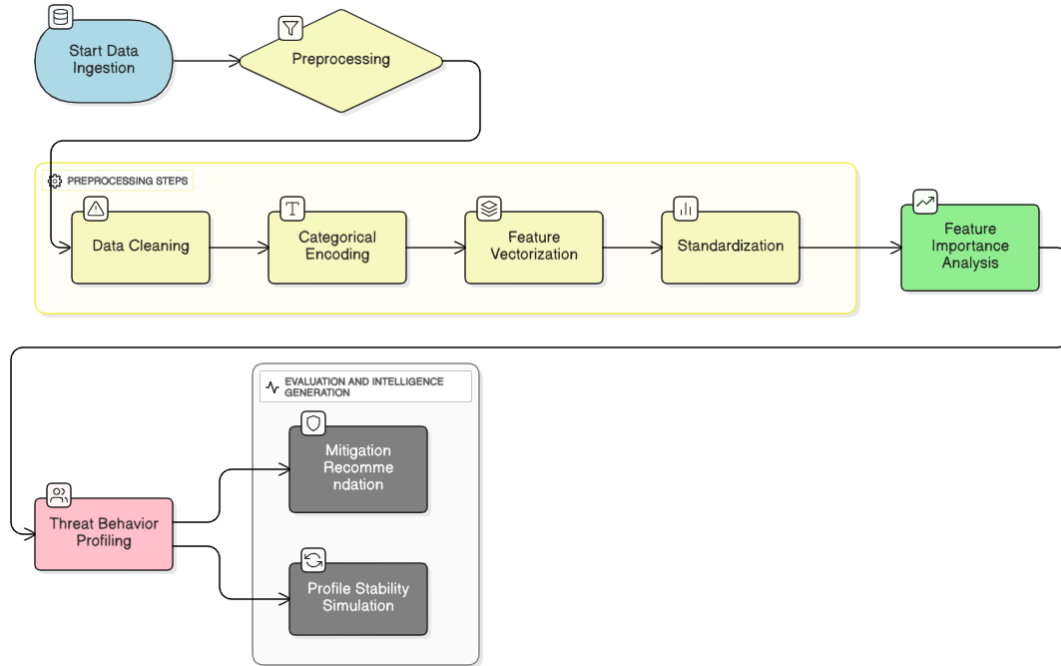


Figure 1: Flow of data through various stages of the analytics pipeline.

The implemented pipeline consists of the following sequential stages:

1. **Data Ingestion from HDFS:** Loading raw CSV or Parquet files into a Spark DataFrame.
2. **Preprocessing:** Cleaning, transforming, and scaling the data for machine learning.
3. **Feature Importance Analysis:** Using a Decision Tree model to identify the most predictive features.
4. **Threat Behavior Profiling:** Applying K-Means clustering to segment attacks into behavioral groups.
5. **Mitigation Recommendation:** Using Alternating Least Squares (ALS) to suggest countermeasures.
6. **Profile Stability Simulation:** Running Monte Carlo simulations to validate cluster robustness.

2.1 Preprocessing

Raw network data is often noisy and unsuitable for direct use in machine learning models. Preprocessing was a critical first step. The following techniques were adopted using PySpark's libraries:

- **Data Cleaning:** Column names were sanitized. Null or infinite values in numeric columns were imputed with zero.
- **Categorical Encoding:** The categorical `label` column was converted into a numerical format using `pyspark.ml.feature.StringIndexer`.
- **Feature Vectorization:** Numeric feature columns were combined into a single vector column using `pyspark.ml.feature.VectorAssembler`.
- **Standardization:** Feature vectors were standardized using `pyspark.ml.feature.StandardScaler` to have a mean of 0 and a standard deviation of 1.

2.2 Building the Model: Threat Profiling

The core of the analysis is the threat profiling engine, which uses unsupervised clustering (`pyspark.ml.clustering.KMeans`) to find distinct behavioral patterns ($k = 3$) within each attack category. Each cluster represents a different variant of the attack, characterized by its centroid, which forms the “fingerprint” of the attack profile.

2.3 Model Evaluation and Comparison

- **Mitigation Recommendation Engine:** A recommendation engine was implemented using the `pyspark.ml.recommendation.ALS` (Alternating Least Squares) algorithm to identify optimal mitigation strategies for different types of cyberattacks. Each identified *threat profile* was treated as a “user,” while the various *mitigation actions* (such as firewall rule changes, IP blocking, or retraining of detection models) were treated as “items.” The model was trained on synthetic data representing the effectiveness score of each mitigation action against a given attack type. After training, the ALS model predicts the top- N mitigation strategies most likely to be effective for a new or unseen threat profile. This enables dynamic and data-driven decision-making for cybersecurity responses, ensuring that similar attack behaviors receive similar or previously successful countermeasures. The system thus helps in automating part of the security operations workflow.
- **Profile Stability (Monte Carlo Simulation):** To assess how consistent and reliable the discovered clusters are, a Monte Carlo simulation was conducted. For each cluster centroid (representing a threat profile), 500 synthetic data points were generated by adding Gaussian noise to simulate slight variations in the attack patterns. These points were then reclassified using the trained K-Means model. The *Stability Score* was defined as the percentage of generated points that remained in the same cluster as their parent centroid. A high stability score (greater than 90%) indicates a strong separation between clusters and suggests that the behavioral patterns identified are both robust and reproducible under data perturbations. Conversely, lower stability indicates overlapping clusters and possible data drift, prompting further tuning or re-clustering.

3 RESULT ANALYSIS

This chapter presents a detailed analysis of the results obtained from both datasets. We first examine the model’s performance on high-level categories and then delve into the granular, specific attack profiles.

3.1 High-Level Profiling on CTU-13 Dataset

The analysis on the CTU-13 dataset served as a crucial baseline, testing the pipeline’s ability to differentiate between the broad categories of ‘ATTACK’ and ‘BENIGN’ traffic. The K-Means algorithm successfully identified three distinct behavioral clusters within each class.

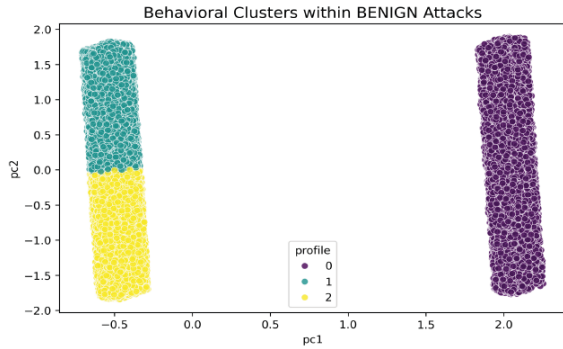


Figure 2: Behavioral clusters within BENIGN traffic (CTU-13).

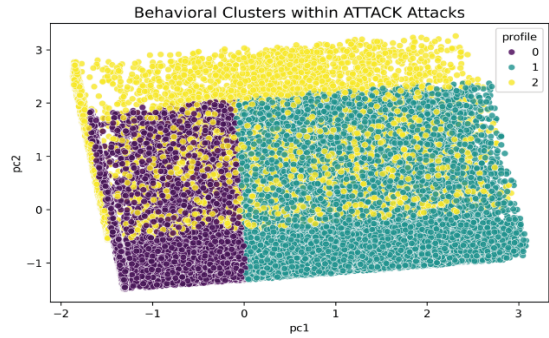


Figure 3: Behavioral clusters within ATTACK traffic (CTU-13).

Table 3: Profile fingerprints for ATTACK traffic (CTU-13).

Feature	Profile 0	Profile 1	Profile 2
Flow Duration	348365.87	2895612.38	1548470.19
Destination Port	16903.90	14139.45	17942.45
Total Fwd Packets	2.13	11.40	6.51

The stability analysis for these broad categories was particularly insightful. While BENIGN Profile 2 showed very high stability (99.20%), the ATTACK profiles exhibited significantly lower scores (e.g., Profile 1 at 34.00%). This result suggests that the generic ‘ATTACK’ label is too broad and encompasses a wide variety of behaviors, making its sub-profiles less consistent and harder to define.

3.2 Granular and Scalable Profiling on CIC-collection

Building on the baseline, the pipeline was executed on the much larger CIC-collection dataset to perform fine-grained analysis of specific, labeled attacks. This demonstrated the system’s true potential for generating high-fidelity intelligence. The model uncovered clearly distinct behavioral clusters within several specific attack categories, as detailed below. Each attack’s profiles are presented on a dedicated page for clarity.

3.2.1 DoS-Hulk Attack Profiles

The DoS-Hulk attack, designed to overwhelm web servers with unique requests, presents three distinct behavioral variants. As seen in the fingerprints below, Profiles 0 and 2 are characterized by extremely long idle times, suggesting they may employ stealthier, intermittent bursts. In contrast, Profile 1 has a significantly shorter idle time and smaller packet sizes, indicating a more aggressive and continuous flooding strategy. This differentiation is critical for tuning rate-limiting and behavioral anomaly detection rules.

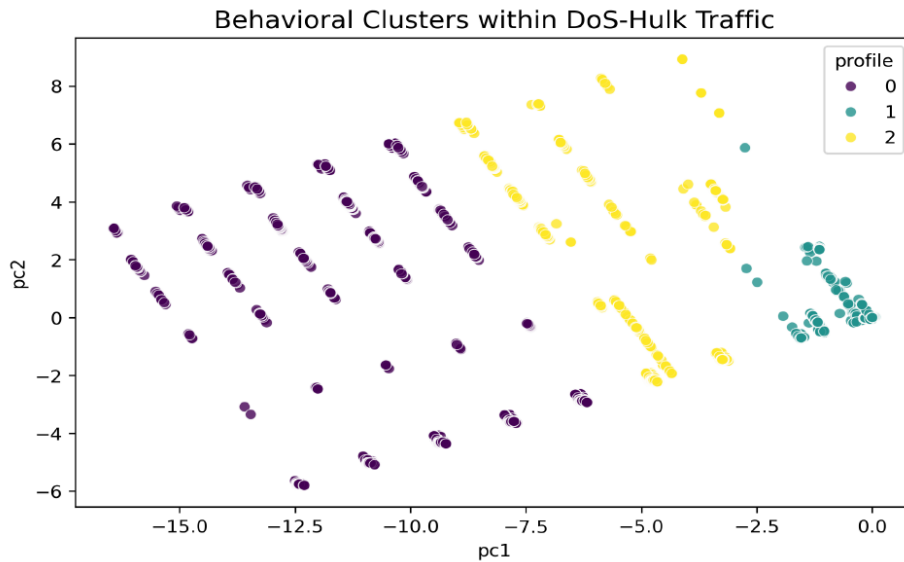


Figure 4: Behavioral Clusters within DoS-Hulk Traffic.

Table 4: Profile Fingerprints for DoS-Hulk Traffic.

Feature	Profile 0	Profile 1	Profile 2
Fwd Packet Length Max	347.01	53.16	346.65
Fwd Packets Length Total	412.81	59.40	431.72
Init Fwd Win Bytes	2,842.34	2,563.17	2,839.26
Total Backward Packets	6.02	0.77	6.53
Fwd Act Data Packets	1.86	0.25	1.89
Bwd Packet Length Max	6,764.27	141.52	4,223.27
Total Fwd Packets	6.68	2.82	6.70
Fwd Iat Total	84,659,248.72	1,119,993.02	83,609,077.47
Idle Min	82,719,187.65	376,585.53	81,763,878.70
Fwd Seg Size Min	26.07	31.06	26.13

3.2.2 DDoS-LOIC-HTTP Attack Profiles

The profiles generated for DDoS attacks using the Low Orbit Ion Cannon (LOIC) tool reveal vastly different operational modes. Profile 1 is the classic high-volume flood, distinguished by an enormous number of forward data packets and an exceptionally long forward inter-arrival time total, indicating a sustained, overwhelming assault. Conversely, Profile 2 exhibits an extremely high minimum idle time, characteristic of a "slow and low" attack designed to evade simple threshold-based detection. Profile 0 represents a more moderate variant.



Figure 5: Behavioral Clusters within DDoS-LOIC-HTTP Traffic.

Table 5: Profile Fingerprints for DDoS-LOIC-HTTP Traffic.

Feature	Profile 0	Profile 1	Profile 2
Fwd Packet Length Max	20.00	32.00	0.02
Fwd Packets Length Total	20.00	4,361,127.14	265.99
Init Fwd Win Bytes	8,192.00	-1.00	1,959.35
Total Backward Packets	4.00	0.00	0.00
Fwd Act Data Packets	1.00	136,284.22	8.31
Bwd Packet Length Max	964.00	0.00	0.00
Total Fwd Packets	3.00	136,285.22	10.31
Fwd Iat Total	559.50	119,332,232.70	22,367,538.53
Idle Min	0.00	0.00	22,035,129.33
Fwd Seg Size Min	20.00	8.00	19.99

3.2.3 DDoS-HOIC Attack Profiles

Analysis of High Orbit Ion Cannon (HOIC) traffic reveals a highly anomalous variant in Profile 1. This profile is defined by zero-valued packet lengths and backward packets, alongside a minimal forward packet count. This suggests it may represent connection-request floods (e.g., SYN floods) or malformed packet attacks rather than volumetric data transfer. In contrast, Profiles 0 and 2 appear to be more conventional, differing primarily in their maximum forward packet length, but are otherwise nearly identical.

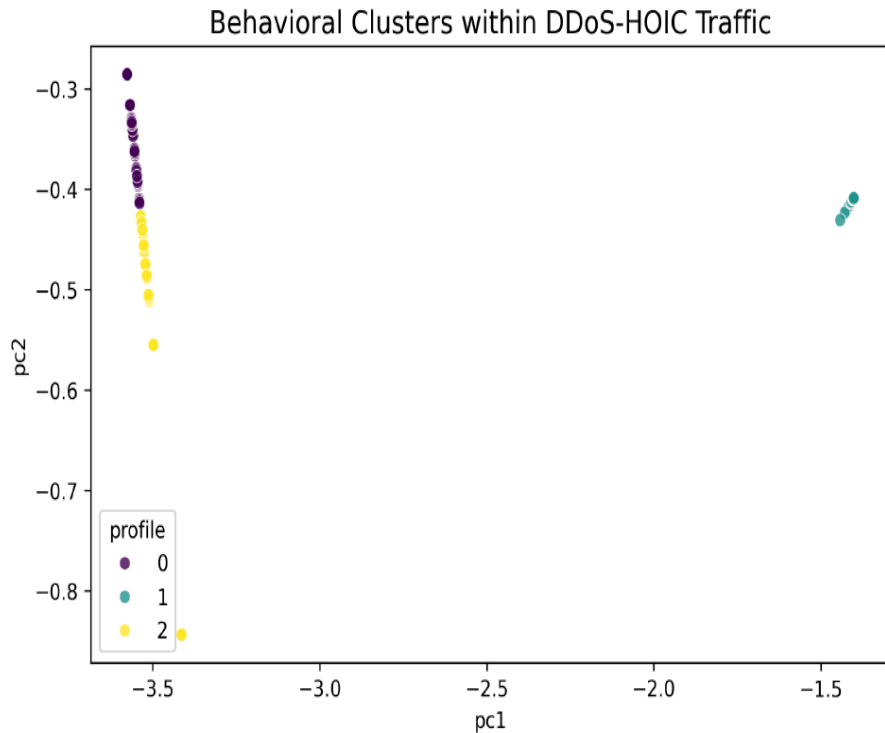


Figure 6: Behavioral Clusters within DDoS-HOIC Traffic.

Table 6: Profile Fingerprints for DDoS-HOIC Traffic.

Feature	Profile 0	Profile 1	Profile 2
Fwd Packet Length Max	320.21	0.00	267.67
Fwd Packets Length Total	320.21	0.00	267.67
Init Fwd Win Bytes	65,535.00	32,738.00	65,535.00
Total Backward Packets	4.00	0.00	4.00
Fwd Act Data Packets	1.00	0.00	1.00
Bwd Packet Length Max	935.00	0.00	935.00
Total Fwd Packets	3.00	2.00	3.00
Fwd Iat Total	8,904.11	18,815.72	8,886.83
Idle Min	0.00	0.00	0.00
Fwd Seg Size Min	20.00	20.00	20.00

3.2.4 Botnet Attack Profiles

The three identified botnet profiles point to different functionalities within the botnet. Profile 1 is the most alarming, characterized by an exceptionally large maximum forward packet length and total packet volume. This pattern is a strong indicator of large-scale data exfiltration. Profile 2 also shows significant data transfer, though less intense. Profile 0, with much smaller packet sizes and volumes, likely represents command-and-control (C2) communication or heartbeat signals, which are essential for maintaining the botnet's operation.

Detailed Profile: Botnet

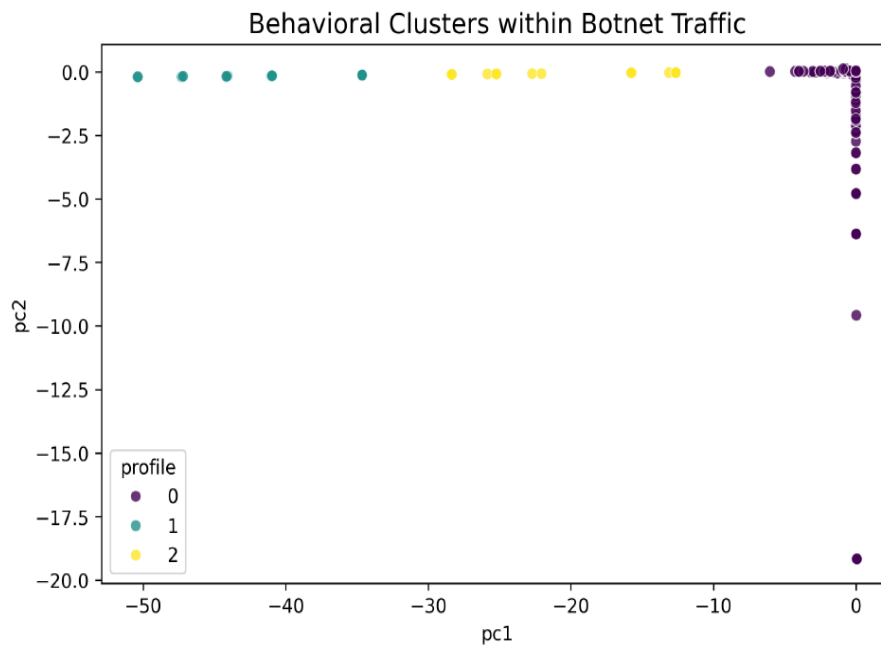


Figure 7: Behavioral Clusters within Botnet Traffic.

Table 7: Profile Fingerprints for Botnet Traffic.

Feature	Profile 0	Profile 1	Profile 2
Fwd Packet Length Max	312.97	19,577.27	8,930.73
Fwd Packets Length Total	313.58	151,057.59	77,051.86
Init Fwd Win Bytes	8,108.06	8,192.00	8,192.00
Total Backward Packets	3.99	64.41	35.50
Fwd Act Data Packets	1.02	30.64	22.41
Bwd Packet Length Max	116.52	128.00	128.00
Total Fwd Packets	3.12	31.64	23.41
Fwd Iat Total	139,325.50	114,396.00	117,809.95
Idle Min	48,160.67	0.00	0.00
Fwd Seg Size Min	19.99	20.00	20.00

3.2.5 DDoS Attack Profiles (Generic)

The profiling of generic DDoS traffic showcases the engine’s ability to separate attacks based on their intensity and structure, visualized as distinct vertical bands. Profiles 0 and 2 are nearly identical high-volume attacks, differing only slightly in their backward packet and idle time characteristics, suggesting they may originate from the same tool with minor configuration changes. Profile 1 represents a low-intensity variant, with minimal packet counts and backward traffic, likely aimed at evading basic volumetric detection systems while still contributing to the overall attack.

Detailed Profile: DDoS

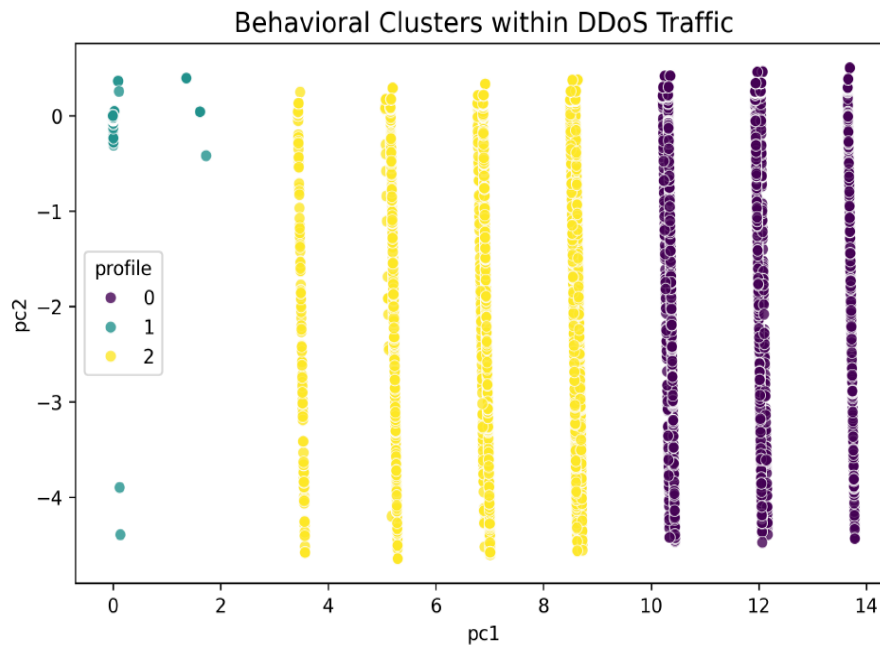


Figure 8: Behavioral Clusters within Generic DDoS Traffic.

Table 8: Profile Fingerprints for Generic DDoS Traffic.

Feature	Profile 0	Profile 1	Profile 2
Fwd Packet Length Max	20.00	6.17	20.00
Fwd Packets Length Total	34.68	27.53	34.66
Init Fwd Win Bytes	5,984.47	282.24	5,967.76
Total Backward Packets	4.37	0.01	5.54
Fwd Act Data Packets	3.16	3.52	3.16
Bwd Packet Length Max	10,011.34	1.84	5,696.93
Total Fwd Packets	4.45	4.52	4.44
Fwd Iat Total	22,513,549.97	6,227,077.04	22,367,443.07
Idle Min	10,979,340.47	5,206,600.06	10,833,550.96
Fwd Seg Size Min	20.00	20.01	20.00

3.2.6 DDoS-NTP Attack Profiles

The profiles for DDoS attacks leveraging the Network Time Protocol (NTP) likely correspond to an amplification attack. Profiles 0 and 1 show a nearly identical, large forward packet length, consistent with spoofed requests sent to NTP servers. Profile 2 uses a smaller request packet. The most striking feature is the massive negative value for ‘Fwd Seg Size Min’ in Profile 1, which points to a data processing anomaly or a specific, malformed packet structure designed to trigger an unusual response. The lack of backward packets in Profiles 1 and 2 confirms the spoofed, one-way nature of the attack traffic.

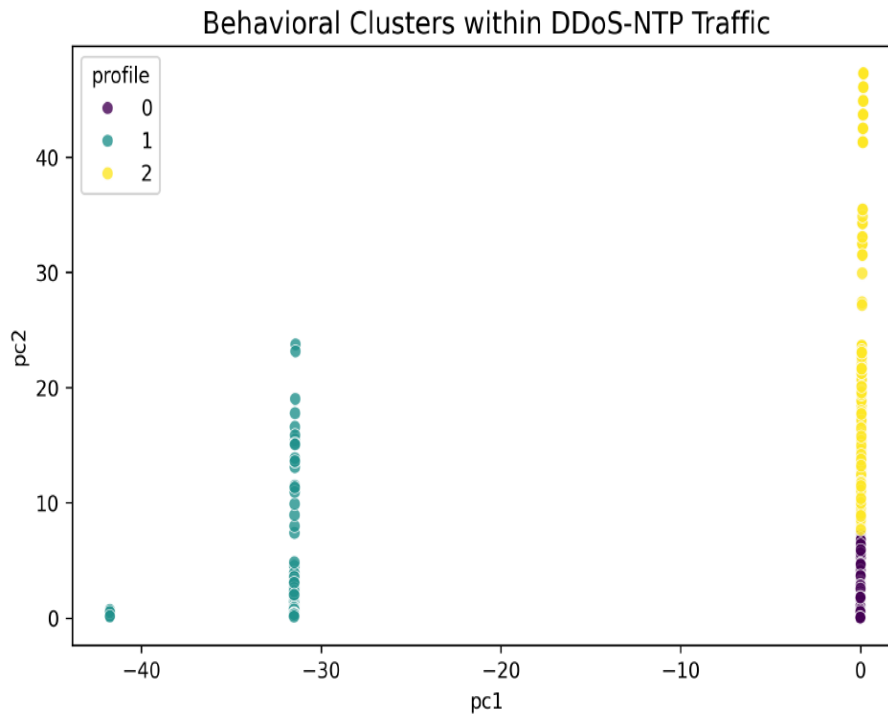


Figure 9: Behavioral Clusters within DDoS-NTP Traffic.

Table 9: Profile Fingerprints for DDoS-NTP Traffic.

Feature	Profile 0	Profile 1	Profile 2
Fwd Packet Length Max	438.77	438.84	421.23
Fwd Packets Length Total	29,826.93	35,155.69	2,889.00
Init Fwd Win Bytes	53.68	-1.00	0.10
Total Backward Packets	0.02	0.00	0.00
Fwd Act Data Packets	67.19	79.29	6.34
Bwd Packet Length Max	0.81	0.00	0.00
Total Fwd Packets	68.20	80.29	7.34
Fwd Iat Total	14,229.62	17,858.59	6.65
Idle Min	1,828.52	0.00	0.00
Fwd Seg Size Min	14.02	-1,064,105,208.05	24.69

3.3 Comparative Analysis and Actionable Intelligence

The most compelling results emerge from the stability analysis, which quantifies the robustness of the discovered profiles. This analysis was performed on both datasets to contrast the effectiveness of profiling on broad versus specific attack labels.

First, the stability analysis on the CTU-13 dataset (Table 10) reveals significant inconsistencies. While one benign profile achieved high stability (99.20

Table 10: Profile Stability Scores for Broad Categories (CTU-13).

Attack Type	Profile ID	Stability Score (%)
BENIGN	Profile 0	44.20
BENIGN	Profile 1	50.40
BENIGN	Profile 2	99.20
ATTACK	Profile 0	50.20
ATTACK	Profile 1	34.00
ATTACK	Profile 2	37.20

In stark contrast, the results from the granular CIC-collection dataset are exceptionally strong. To provide a clear overview, Table 11 presents the **average stability score calculated across all three discovered profiles** for each specific attack type. The vast majority of attack types show near-perfect stability, proving the system can identify and validate extremely consistent and well-defined threat signatures. Even for DDoS-HOIC, which has a lower average, the analysis revealed one profile with 100% stability, demonstrating the engine’s ability to find at least one highly reliable signature within a more diverse attack category.

Table 11: Average Profile Stability Scores for Granular Attacks (CIC-collection).

Attack Type	Average Stability Score (%)
DDoS-LOIC-HTTP	99.80
DoS-Hulk	100.00
DDoS-HOIC	73.67
Botnet	100.00
DDoS (Generic)	100.00
DDoS-NTP	100.00

Comparing Models and Datasets: The CTU-13 analysis demonstrates that profiling on broad, generic labels like ‘ATTACK’ yields profiles with low stability and consistency. In contrast, the CIC-collection analysis shows that when the same model is applied to specific, well-labeled attack data, it discovers sub-profiles that are extremely stable and behaviorally distinct. This comparison underscores a key conclusion: **the quality and granularity of data labels are paramount for effective automated threat profiling.**

Monte Carlo Stress Test Graph (CTU-13 Dataset)

This section displays the results of the Monte Carlo stability stress test for dataset-1. This test evaluates the robustness of the identified threat profiles by introducing random noise to the data points and observing if they remain within their original cluster. The graphs show the stability score (the percentage of points that remain correctly classified) as the noise level increases. High, stable scores indicate well-defined and reliable cluster profiles.

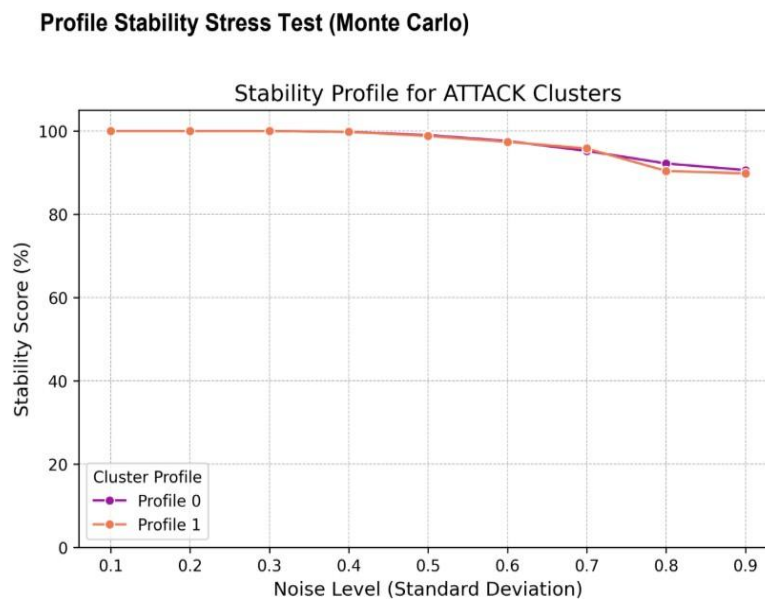


Figure 10: Stability Profile for ATTACK Clusters.

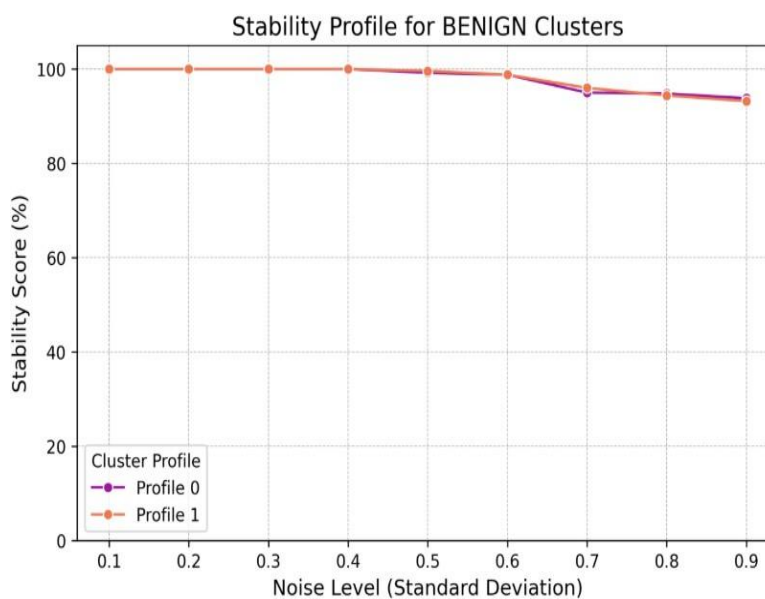


Figure 11: Stability Profile for BENIGN Clusters.

Monte Carlo Stress Test (CIC Collection - Part 1)

The following graphs from the CIC Collection dataset show the stability of profiles for specific, granular attack types. The results for DoS-Hulk and DDoS-LOIC-HTTP attacks are exceptionally strong, with average stability scores of 100

Profile Stability Stress Test (Monte Carlo)

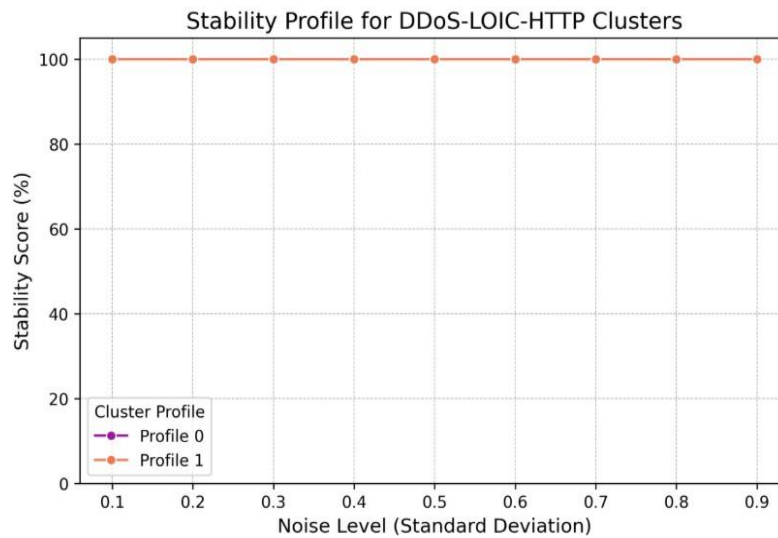


Figure 12: Stability for DDoS-LOIC-HTTP.

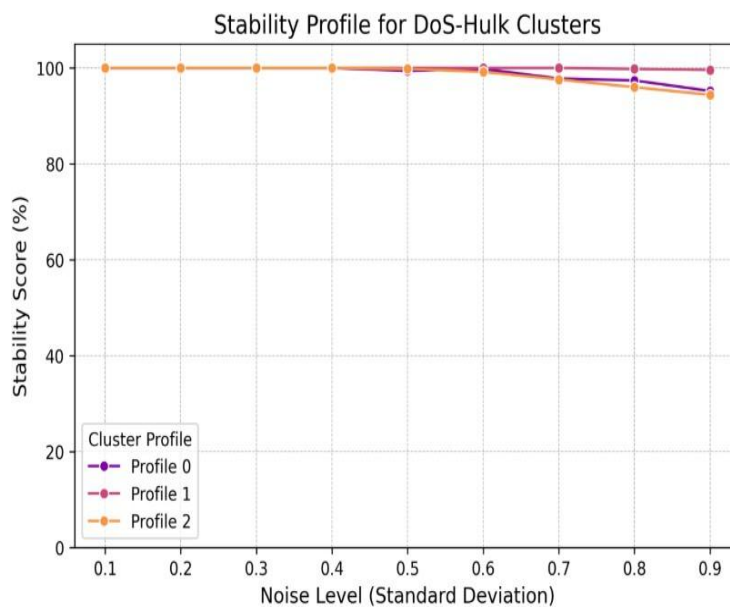


Figure 13: Stability for DoS-Hulk.

Monte Carlo Stress Test (CIC Collection - Part 2)

This page contrasts two different attack types with distinct stability characteristics. The Botnet profiles demonstrate perfect (100%) stability, suggesting a highly uniform and repeatable attack signature, which is characteristic of a centrally controlled network where all bots execute identical commands. In contrast, the DDoS-HOIC attack has a lower average stability of 73.67

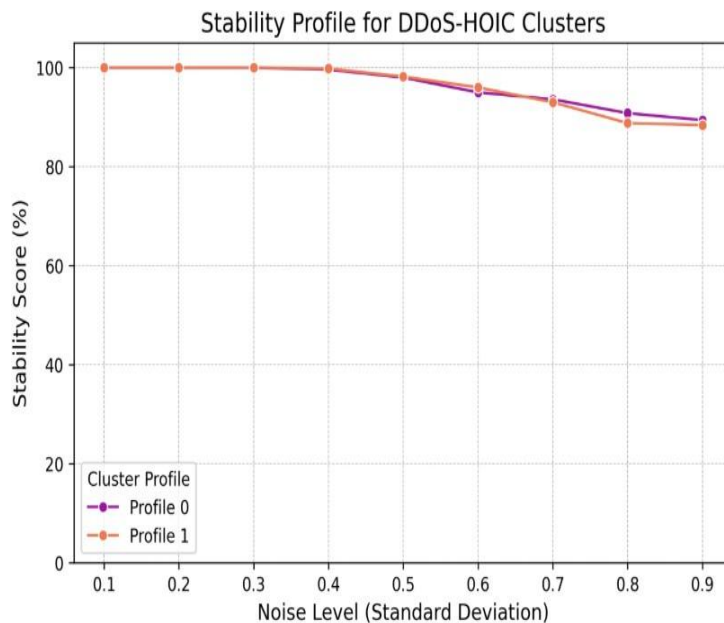


Figure 14: Stability for DDoS-HOIC.

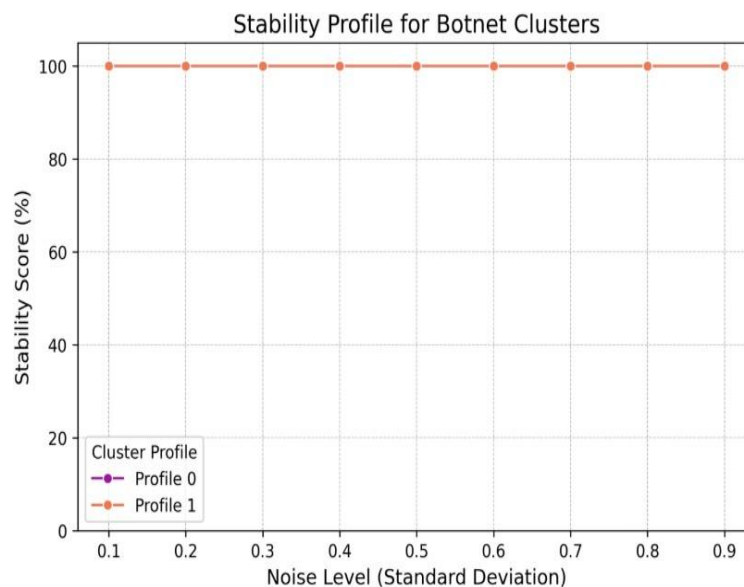


Figure 15: Stability for Botnet.

Monte Carlo Stress Test (CIC Collection - Part 3)

The final set of profiles for the generic DDoS and DDoS-NTP attacks both show perfect 100

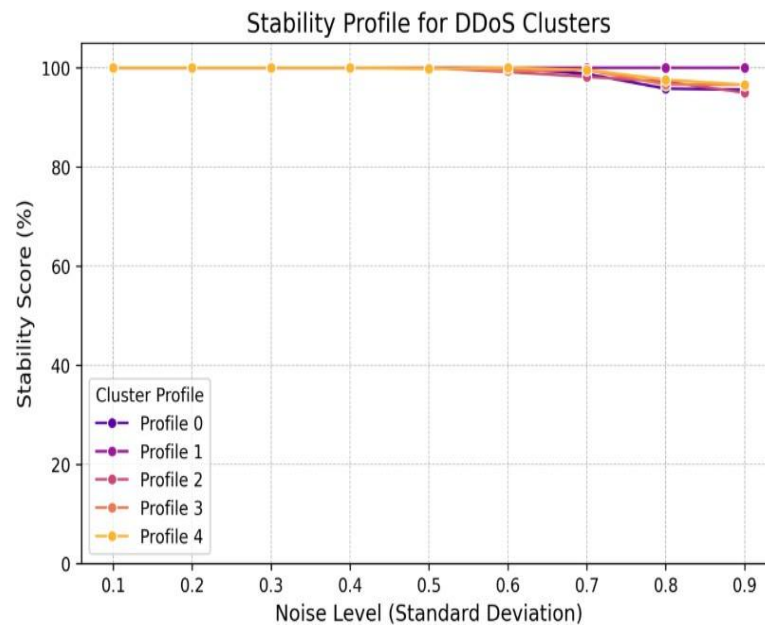


Figure 16: Stability for Generic DDoS.

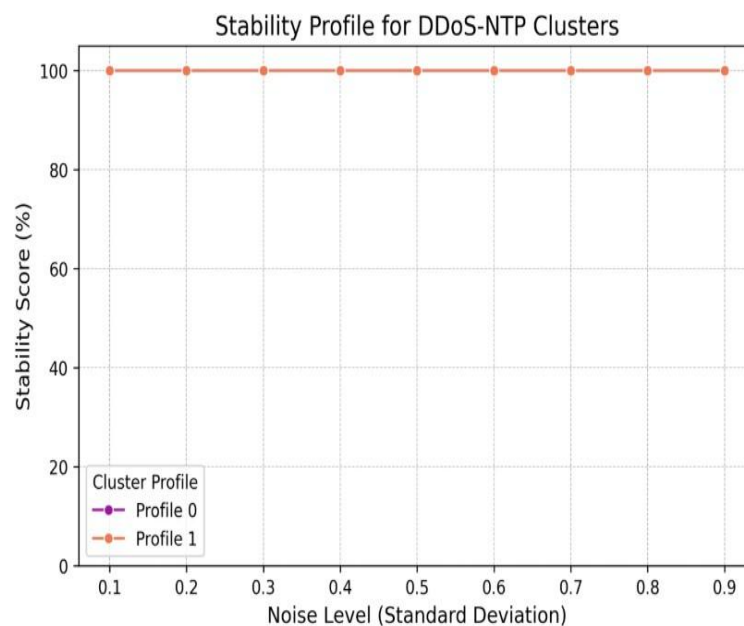


Figure 17: Stability for DDoS-NTP.

4 CONCLUSION

This project successfully designed and implemented a scalable, multi-stage threat intelligence pipeline using Apache Spark. The system demonstrated its ability to process millions of network records to automatically discover distinct behavioral profiles within various attack categories. The results confirm that different variants of the same attack type can have significantly different network fingerprints, and this system provides the means to detect and characterize them at scale. The high stability scores achieved on granular attack labels from the CIC-collection dataset validate the robustness of this approach, proving its potential as a powerful tool for automating threat analysis in a Security Operations Center (SOC).

The primary limitations of this project stem from the dataset and the environment. The mitigation recommendation engine, while functionally sound, was trained on very homogeneous data due to the lack of publicly available diverse datasets that map specific network threat profiles to verified incident response outcomes. The effectiveness of this engine in a real-world scenario is therefore hypothetical and would require extensive training on historical security logs. Furthermore, the analysis was conducted on a simulated cluster, which does not fully account for real-world complexities like network latency and hardware failures that can impact performance at scale. The choice of K-Means also requires the number of clusters (k) to be predefined, which can be a subjective and challenging task for unknown or evolving threats.

Future work should focus on enhancing the pipeline's automation, intelligence, and real-world applicability. A crucial next step is to integrate the system with a real-time data ingestion framework, such as **Apache Kafka and Spark Streaming**, to enable near-real-time threat profiling. The profiling methodology could also be extended to detect **zero-day threats** by developing an anomaly detection module that flags new clusters of traffic that do not conform to any known benign or malicious profiles. To improve model performance, future iterations could incorporate **automated hyperparameter tuning** and explore more advanced clustering algorithms like **DBSCAN**, which can identify arbitrarily shaped clusters without requiring a predefined cluster count. Finally, integrating **Explainable AI (XAI)** techniques would be invaluable, providing human-readable justifications for why a particular activity was flagged, thereby empowering security analysts to make faster and more confident decisions.

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